SPE-227499-MS

OQEP Digitalization Case: Automation of Real-Time Water Cut Calculation Using High Frequency Sensor Data and Well Test Data of Digital Oil Field System

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This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

Abstract

Objectives/Scope: Automate the calculation of Near Real-time Water cut (RTWC) for ESP wells using High frequency sensor data (HFSD) and well tests data (WTD) available in Digital Oil field (DOF) platform, utilizing Digital twins benefitting OQEP's DOF and Production monitoring teams. A step change in the Digital Oil field Arena towards virtual metering, workflow integration with production operations solutions, improving DOF system performance, maintaining evergreen updated live data, integration of results for cross-domain analysis, integration with Machine Learning & Artificial Intelligence models (ML/AI).

Methods, Procedures, Process: The method for real-time water-cut estimation in ESP wells leverages field data from surface and downhole sensors, well PVT data, and physics-based models. By tracking changes in parameters like Pump Discharge Pressure (PDP) and Well Head Pressure (WHP), the system calculates water-cut using standard petroleum engineering equations. Integrated into the DOF field surveillance application, it automatically performs scenario simulations and sensitivity analyses to ensure accurate results. The RTWC data is visualized within the Well Management Visualization System (WMVS), aiding engineers in decision-making and intervention planning, leading to significant time savings, optimization, and enhanced operational visibility.

Results, Observations, Conclusions: The implementation of the RTWC estimation method for ESP wells has delivered significant operational improvements for OQEP. Key results include a 30% time saving for engineers in tracking accurate RTWC, resulting in more efficient well management. Additionally, the method has contributed to a 5% improvement in field factor calculations and a 10-20% reduction in well-test OPEX for ESP wells, indicating cost savings and operational efficiencies. Furthermore, the system has enabled 100% digital transparency into ESP operations, enhancing real-time monitoring and decision-making. The water-cut readings generated by the system, when coupled with the WMVS, have been instrumental in helping engineers evaluate phase-wise production, correct misleading well test measurements, and track water production in real-time. These capabilities provide engineers with valuable insights to make informed decisions on well interventions and optimizations. From an operational

standpoint, this method has proven effective despite being a relatively small module within the broader well optimization process. Its integration with AI-based predictive models is expected to strengthen the ability to detect potential well failures early, improving planning and minimizing disruptions. Overall, the method's automation and integration with existing systems have significantly enhanced field productivity and decision-making capabilities, marking a critical milestone for OQEP's digital oil field strategy.

Novel/Additive Information: Accurate water-cut estimation is crucial for optimizing ESP well performance, as it helps differentiate between oil and water phases in production. This real-time calculation allows engineers to detect production anomalies early, adjust operational parameters, and enhance well efficiency, ultimately improving reservoir management and reducing operational costs.

Introduction

Water cut (WC) is a critical parameter in the optimization of production processes, reservoir management, and the accurate tracking of reserve positions. The importance of WC lies in its direct influence on understanding the fluid composition being produced, which is essential for making informed decisions regarding production efficiency, reservoir performance, and the financial impact of maintaining oil-water separation processes. Since the frequency of well testing is often limited due to operational constraints and costs, engineers are often left with uncertainty regarding how the WC has fluctuated between tests. This gap in data creates challenges, especially when it comes to assessing well behavior and determining the changes in fluid dynamics over time. Whenever a new well test is conducted, production engineers are forced to rely on the WC value provided, often without any robust backup measurement or calculative method to cross-check its accuracy. This scenario becomes particularly problematic when the well in question has undergone significant changes in its downhole conditions, such as fluctuating pressures or other operational adjustments.

For wells with relatively stable behaviors, particularly those employing ESP (Electric Submersible Pumps), trend analysis can help estimate the WC by identifying patterns in production data. However, for wells that exhibit unstable behavior or show significant variability in downhole pressures, accurately estimating the WC and forecasting potential oil deferments becomes increasingly difficult. These variations complicate the task of monitoring water cut and significantly reduce the accuracy of any estimates made using traditional methods. Moreover, the ability to estimate the water cut with higher accuracy is not only valuable for monitoring the well's performance but also essential for optimizing the oil-water separation process, ultimately helping to improve overall production efficiency.

The proposed method to calculate WC more accurately involves leveraging real-time data from ESP sensors, such as Pump Discharge Pressure (P_{PDP}), Pump Intake Pressure (P_{PIP}), and Tubing Head Pressure (P_{THP}), which are continuously recorded and stored in a surveillance and monitoring tool. This real-time data, combined with well test data, provides a more holistic view of well performance. The daily averaged data from the ESP sensors, alongside critical well test parameters like flow rates, tubing head pressure, and other required data points, can be used to calculate WC for all wells automatically and with greater accuracy.

Initially, the calculations were based on simple mathematical equations (from Equation 1 to Equation 12), but the process has since been integrated into the Well Management Visualization System (WMVS) platform developed by the DOF team. This platform enhances the accuracy and efficiency of WC calculations across different ranges of values, allowing for automatic updates and real-time validation. The integration of a standard Application Programming Interface (API) enables seamless, on-the-fly calculations or scheduled processing according to user needs. Additionally, the data is stored in a Structured Query Language (SQL) database, facilitating real-time plotting, analysis, and validation.

Furthermore, this methodology allows for daily WC estimates to be incorporated in inferred rates derived from physics-based well models or digital twins. This conceptual approach not only enhances the precision of water cut predictions but also helps build confidence in the data, supporting better decision-making in terms of reservoir management and production optimization. By providing continuous, real-time estimates of WC, this system significantly improves the accuracy of well performance tracking, thereby enhancing overall operational efficiency.

Methodology Basis

Single phase incompressible liquid flow

ESP wells with minimal free gas, when equipped with a highly efficient downhole separator, can effectively remove free gas before it reaches the ESP Bottom Hole Assembly (BHA). In such cases, NODAL analysis can be performed using straightforward hand calculations. As illustrated in Figure 1, pressure readings are taken at various depths along the tubing string and annulus string. In the absence of gas in the ESP production stream, the well is assumed to produce a stabilized incompressible liquid rate, which is defined by the Flowing Bottom Hole Pressure (FBHP) and constant Tubing Head Pressure (THP) based on the well's Inflow Performance Relationship (IPR).

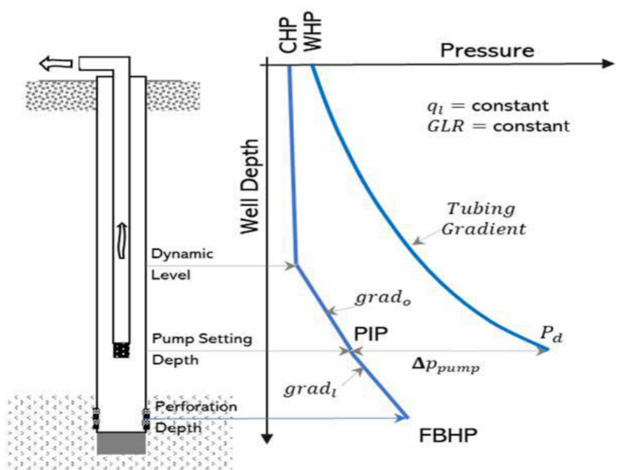


Figure 1—Pressure Distribution in an Electrical Submersible Pump Installation (Takacs, 2018)

To estimate the inferred rates, Physics-based well models were developed utilizing advanced platforms like Physics Based Well Modelling and Digital Oil Field. These models integrate key data such as PVT data, well completion information, and well test results to provide accurate estimates of production rates and other critical performance metrics. The near RTWC generated by this method is also integrated with physics based well models to sync-in the rates estimation.

WC of ESP well can be easily diagnosed by following equations given. It is important to ensure that simple logic is followed during the diagnoses stage (MLF-81). This logic was initially created using a VBA Macro to gain an understanding of the results for wells with higher water cut.

Outflow pressure P_{PDP} equals to sum of

- The pressure required to overcome depth and gravity (P_G)
- The pressure required to overcome the surface system pressure. (P_{THP})
- The pressure required to overcome the flowing friction (P_f)

$$P_{PDP} = P_{THP} + P_G + P_f \quad (1)$$

Where,

P_{PDP} = Pressure at Pump Discharge OR Outflow pressure (psi)
P_{THP} = Pressure at Well Head (psi)
P_G = Pressure due to Height of Liquid Column (psi) = Depth (mMD) X G_{AVG}
P_f = Pressure loss due to Tubing Friction (psi)

The API gravity scale is an anticipated but well-established method for measuring density using a hydrometer and data for this and is readily available in PVT dataset. The unit of measure is 'degrees' API. Below Equation 2 can be used to calculate SG_{OIL} from °API gravity:

$$SG_{OIL} = \frac{141.5}{131.5 + {}^\circ API} \tag{2}$$

Where,

SG_{OIL} = Specific Gravity of Oil
°API = Oil gravity

Specific gravity of Water (SG_{H2O}) and Initial Water Cut (WC_i) are readily available in PVT dataset and Well tests.

Using all the above calculated and available data; Specific Gravity of Produced Liquid can be calculated using Equation 3 and Average Fluid Gradient with Equation 4.

$$SG_{AVG} = \left(SG_{H20} X\, WC_i\right) + \left(SG_{0IL} X\left(1 - WC_i\right)\right) \tag{3}$$

$$G_{AVG} = G_W + SG_{AVG} \tag{4}$$

Substitute SG_{AVG} in Equation 4 with Equation 3

$$G_{AVG} = G_W + \left(\left(SG_{H20} X\, WC_i\right) + \left(SG_{0IL} X\left(1 - WC_i\right)\right)\right) \tag{5}$$

Where,

G_{AVG} = Average Fluid Gradient
G_W = Pure Water Gradient = 0.433 psi/ft
SG_{AVG} = Average Specific Gravity
SG_{H2O} = Specific Gravity of Water (Fixed Value from PVT Data)
SG_{OIL} = Specific Gravity of Oil (Calculated using Equation 2)
WC_i = Initial Water Cut from 1st Well test data / Starting datapoint

Friction Component

ESP pump depth, tubing size, and flow rates, all influence the tubing friction factors. The ESP pump depth and tubing size are set typically at 6,000 feet and 3.5 inches, respectively. For the friction factors, there are two models that can be made. The first uses a set pressure, such as 50 psig for flow rates below 1,000 BFPD, 150 psig for flow rates between 1,000 and 3,000 BFPD, and 300 psig for flow rates over 3,000 BFPD. This figure will vary if different tubing sizes are used (SPE-211407-MS). Evaluation of this model was done during our analysis but after careful considerations we progressed with second model. Second model, it is more accurate when utilizing Hazen - William's calculation, although flow rate is a necessary variable.

$$F = 2.083 \frac{\left(\frac{100}{C}\right)^{1.85} \left(\frac{Q}{34.3}\right)^{1.85}}{ID^{4.8655}} \tag{6}$$

Where,

F = Tubing Pipe Friction, psi/1000 ft
C = Hazzen - William Coefficient, C=120 for new tubing and C=95 for used tubing
Q = Flow Rate, BFPD
ID = Inner Diameter of Tubing, inch (From Well completion data)

The chart in Figure 2 can also be used for finding out 'F'.

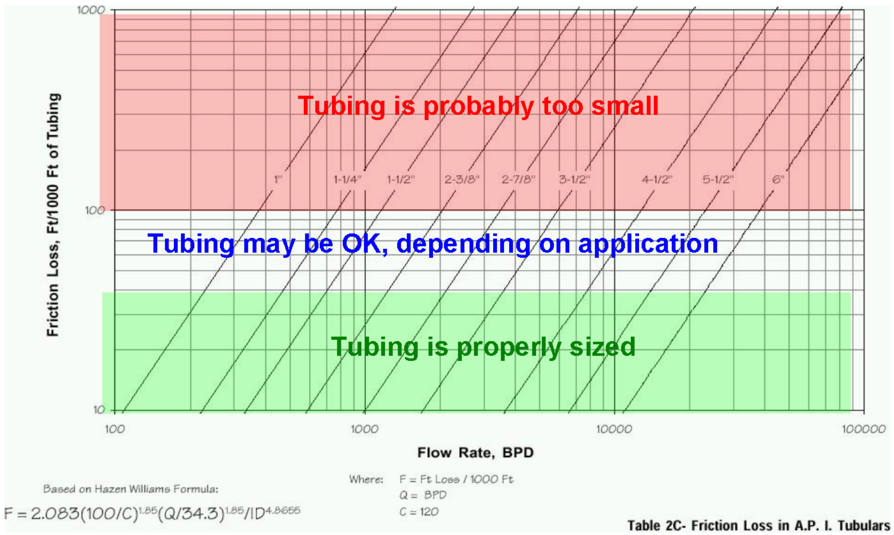


Figure 2—Hazen & Williams Chart for Tubing Sizes (Table 2C – Friction Loss in A.P.I. Tubulars)

To find the Total Friction HEAD (in distance), multiply Tubing Pipe Friction (F) by the total measured length of the tubing:

$$H_{FRICTION} = FXL_{TUBING} \tag{7}$$

Where,
$H_{FRICTION}$ = Head (distance)
L_{TUBING} = Measured Length/Depth of tubing, ft MD
And to find the Pressure loss due to Tubing friction (P_f), conversion of the Friction Head ($H_{FRICTION}$) to Pressure using the Average Fluid Gradient (G_{AVG}) below Equation 8 can be used.

$$P_f = H_{FRICTION} \; X \; G_{AVG} \tag{8}$$

The Real Time data for P_{PDP} and P_{THP} is readily available with WMVS. Using this data along with P_f calculated using Equation 8 the Pressure due to Height of Liquid Column (P_G) can be calculated by rearranging Equation 1 as below.

$$P_G = P_{PDP} - P_{THP} - P_f \tag{9}$$

At this point we have a good estimate of Pressure's and using this data the Average Fluid Gradient (G_{AVG}) and Average Specific Gravity (SG_{AVG}) can be estimated using Equation 10 and Equation 11 respectively.

$$G_{AVG} = \frac{P_G}{ftTVD} \tag{10}$$

At this point iterative calculation is performed from Equation 8 to Equation 10 with an objective to get a close match on G_{AVG} as the first value was calculated based on Initial Water Cut from first Well-test (WC_i).

Post 3 to 5 iterations when G_{AVG} is constant Average Specific Gravity (SG_{AVG}) can be calculated using Equation 11.

$$SG_{AVG} = \frac{G_{AVG}}{G_W} \tag{11}$$

Where,

SG_{AVG} = Average Specific Gravity
G_{AVG} = Average Fluid Gradient <u>after final iterative calculation</u>
G_W = Pure Water Gradient = 0.433 psi/ft

Once the Average Specific Gravity of Produced Liquid was calculated we used this to estimate the Water Cut (WC) rearranging Equation 5 and other rearranged equations as below Equation 12.

$$WC = \frac{SG_{AVG} - SG_{OIL}}{SG_{H2O} - SG_{OIL}} \tag{12}$$

Figure 3 shows the steps involved in calculation of Water Cut (WC) using Real time, Well completion, PVT datasets, etc., from all above-mentioned equations.

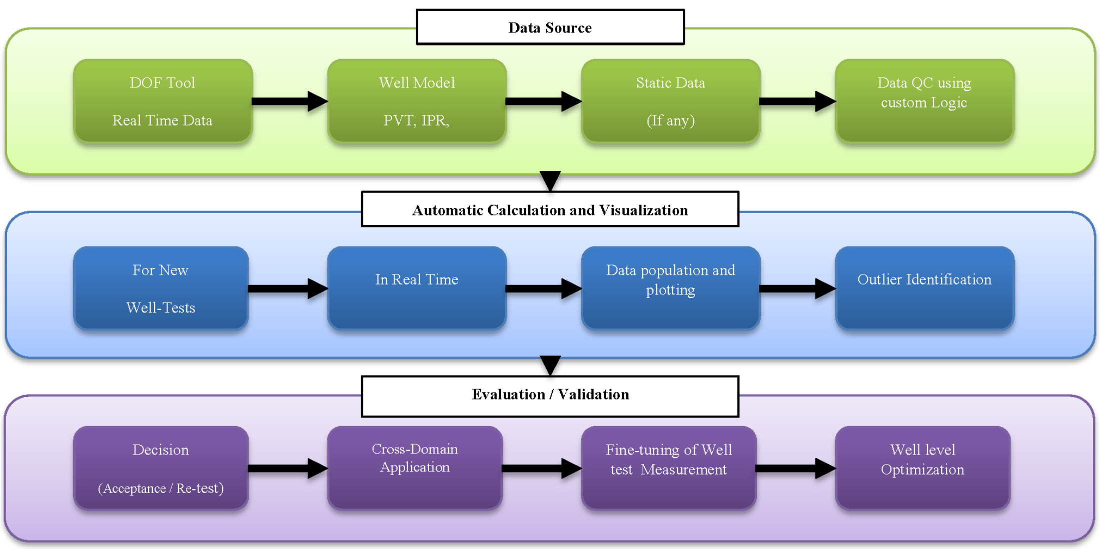


Figure 3—Method Steps for Estimation of Water Cut

While the initial tool developed using a VBA macro gave encouraging results in initial testing, it was limited both in functionality and range by some basic faults. For one, the tool was optimized to perform the calculations with reasonable accuracy only for wells with a water cut in excess of 70%. For wells with lower values of water cut—and particularly those below the 70% threshold—the predictive strength of the tool fell off significantly, producing erroneous outputs. Second, the presence of gas in the production stream was a major challenge. The tool lacked the capability to properly account for the introduced complexities of gas-liquid interactions, which caused further deviations in the modeled outputs. The tool also was highly sensitive to input data quality. Any inconsistencies or errors in the input data—whether they were due to manual input errors or sensor noise—could propagate through the model and give erroneous results.

To overcome these limitations, we embarked on the use of more advanced algorithms in our Digital Oilfield (DOF) suite of algorithms. These algorithms provided a more robust framework that allowed for greater flexibility and reliability across a broader range of well conditions. One of the key improvements was the addition of dynamic scaling factors, which stabilized the supporting equations and improved the model's handling of variable water cut levels in the various wells.

Furthermore, the new models incorporated emulsion behavior that was identified and quantified through machine learning and model training from past production data. This allowed the system to better simulate water cut results with difficult fluid compositions.

Overall, the migration to our DOF suite was a significant leap forward for us to deal with the issues on a more integrated basis, which translated into improved system efficiency and decision-making.

Case Study

The primary goal of this project was to devise a robust and scalable method to estimate water cut in real time from a heterogeneous set of production wells. In an effort to validate the efficiency and precision of the solution presented, we carried out a comprehensive testing of water cut data for over 300 operating wells, dispersed across different fields areas and reservoir pockets. This big data comprised wells with different production profiles, water cut levels, and operational conditions and hence constituted a complete basis for establishing the versatility and precision of the logical model. This allowed us to validate the performance of the logic under various reservoir and production conditions, including different levels of water cut, the presence of gas, and emulsion characteristics.

In this chapter, we present four cases to demonstrate the application of our leading-edge methodology. The cases were selected to demonstrate both the technical innovation in estimating water cut and the practical advantages gained in terms of quality control, operational insight, and decision support.

Each of the examples demonstrates how the greater accuracy of the water cut estimation has enabled generating tangible value from this activity. With more accurate and timely data, the approach has enabled better decision-making in operations, better monitoring of production, and ultimately better efficiency and optimization of well performance management.

Case 1: Well – A

Since the installation of advanced water cut estimation logic on Well A (Figure 4: Case 1 - Well A), we observed a high rate of convergence between calculated readings and actual water cut values within an evenly consistent range of 2% deviation. This consistency is again transmitted in calculation of inferred rates generated from physics based well models where the calculated WC is utilized as shown. To check the model's reliability and accuracy during the initial deployment period, we conducted rigorous manual water cut sampling and laboratory testing. These tests made sure that the calculated data closely approximated the actual conditions, thereby validating the estimation technique.

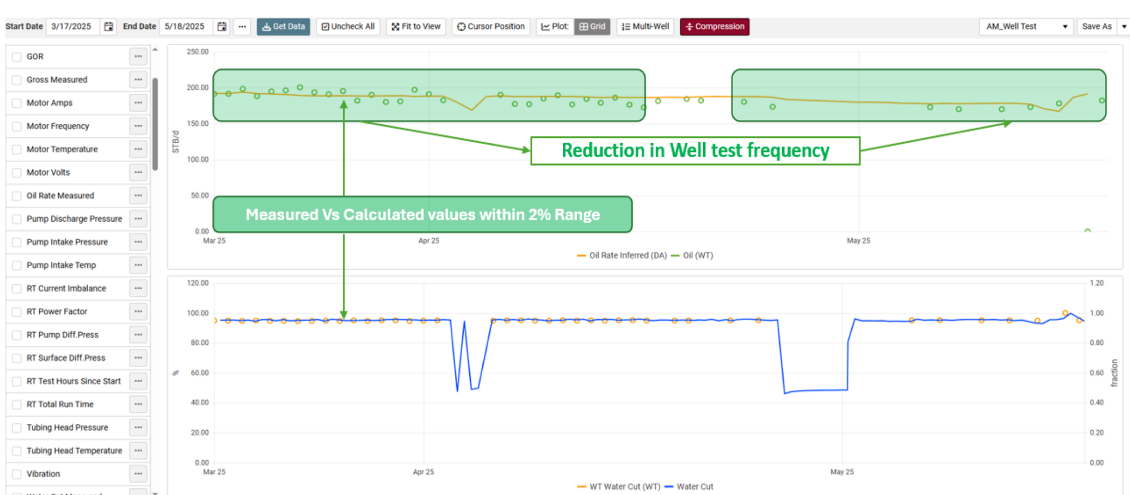


Figure 4—Case 1 - Well A

Because of this validation and as confidence in the methodology increased, the amount of physical well tests—usually needed for direct measurement of water cut—was greatly reduced. Such a paradigm shift has led to a very significant reduction in operational expense (OPEX) to date by approximately 20% in general from all wells. In addition, the continued application of such justification will continue to curtail expenses by reducing the need for recurrent, costly sampling campaigns without diminishing data quality or the capability to monitor production. Such cost savings will be continuous and ongoing.

This case demonstrates the way in which the addition of real-time, accurate water cut estimation not only improves data integrity but also translates to hard dollar returns of operational and financial gain, allowing for better well management and resource utilization.

Case 2: Well – B

For Well B (Figure 5: Case 2 - Well B), the imputed calculation technique for water cut and inferred rates of production has proved particularly valuable with no available actual measured data points being present. As shown in Figure 5, this technique provides for the generation of continuous and reliable estimates of water cut even when physical sampling is not regular or available, to counteract the common issue that exists in many wells with few measurement options.

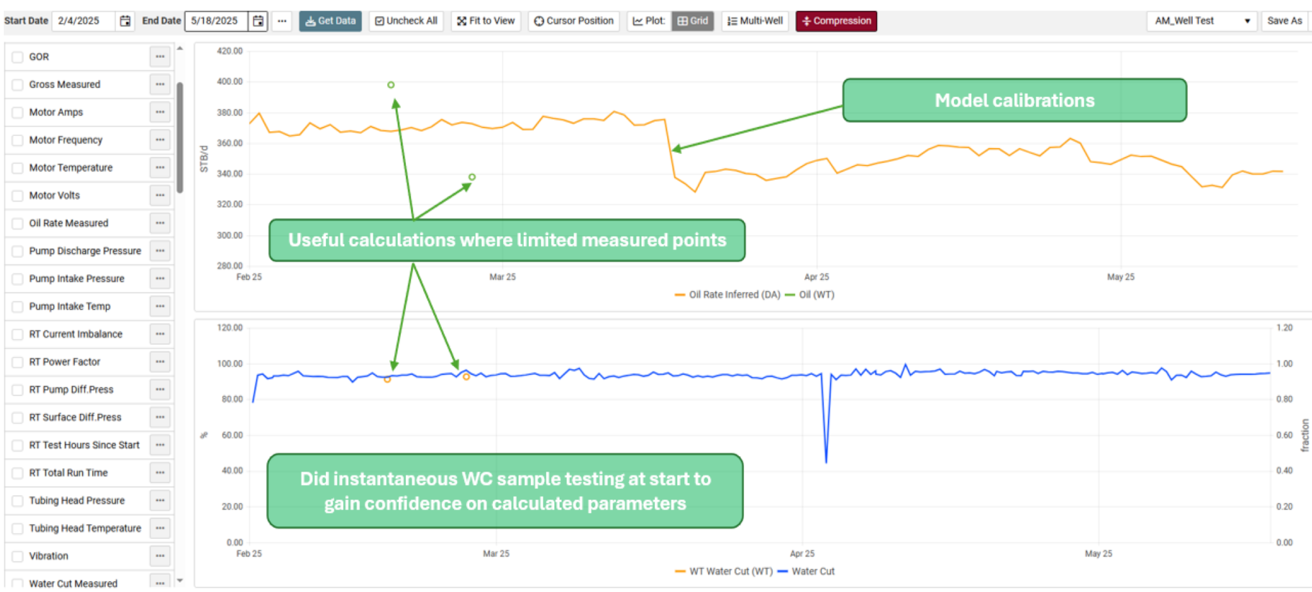


Figure 5—Case 2 - Well B

The system provides production engineers with daily, reliable volumes of computed water cut and rate of production data that significantly benefit them in making timely and fact-based analysis of well performance. The stream of data enables forward-thinking decision-making and can identify trends or anomalies that would be missed between physical readings.

One of the important features of the methodology is its capacity for auto-tuning. As soon as a new valid well test measurement is inserted in the system, the model will be automatically recalibrated to consider this new data, increasing the predictive quality of the model over time. This capability for self-learning allows virtual estimation to agree with real conditions, and thus the tool is adaptive and reliable for ongoing production surveillance.

Overall, Well B illustrates the manner in which virtual water cut estimation is able to overcome data availability limitations while offering operational efficiency and enhanced analytical support to production groups.

Case 3: Well – C

The system is able to track both abrupt and creeping changes in production decline and water cut increase (Figure 6: Case 3 - Well C), showing a composite view of the performance of wells over time. The ability to monitor the changes in production behavior in real-time has provided valuable information on rate interference from offset injectors. By carefully monitoring these parameters, we were able to analyze the decline in production rates and further understand the dynamic well-to-injector or well-to-reservoir interaction. Moreover, the system's advanced analytics have allowed engineers to make data-driven decisions more quickly and accurately. Engineers have, in turn, reduced the time spent analyzing critical production behaviors by more than 30%, significantly improving operational efficiency. This reduction in analysis time has also allowed for more proactive and earlier interventions, optimizing production strategies and optimizing overall reservoir management.

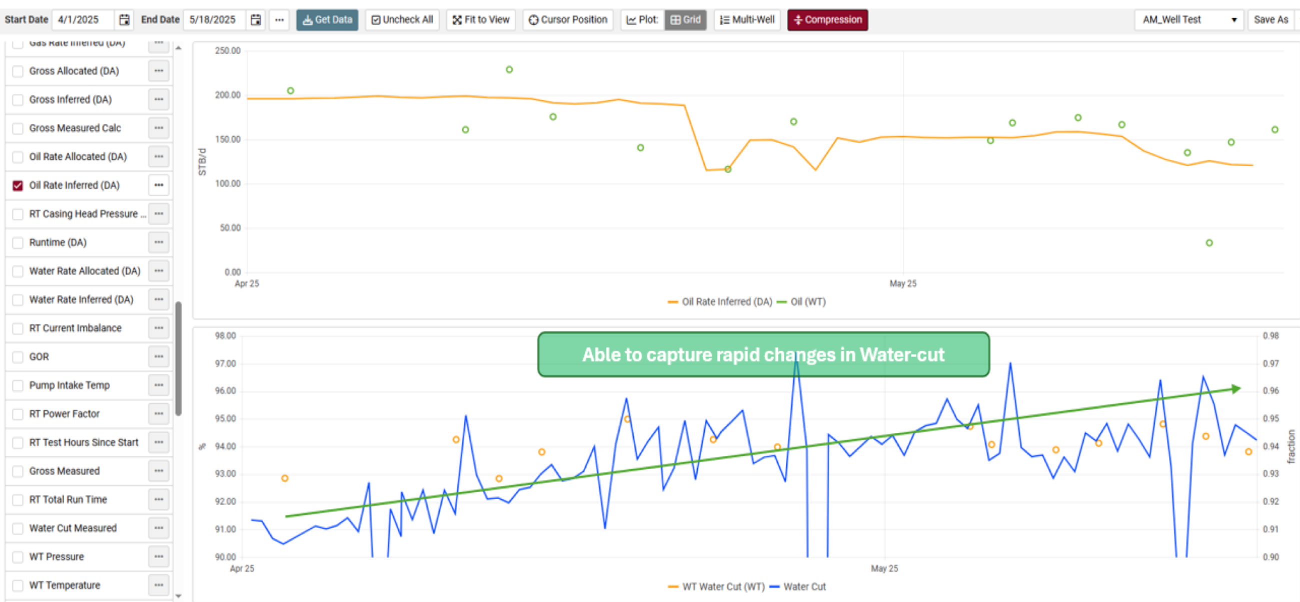


Figure 6—Case 3 - Well C

Case 4: Well – D

The system's logic is such that it trains itself through the continuous changes in input data so that it can automatically evolve and adapt with the new data as it becomes available. This adaptive learning enables the system to stay extremely responsive to any shift in well performance, thus being capable of refining its models and predictions over a period of time. As data keep updating, the system automatically adjusts its algorithms to make more accurate predictions, as one can see in Figure 7: Case 4 - Well D. The system enhances its ability to predict future production trends and potential disruptions in real time with the help of recalibration. This self-learning feature reduces the need for manual recalibration and allows the system to stay aligned with the latest reservoir conditions. Moreover, it helps engineers save time on model recalculation, streamline working processes, and provide more consistent, actionable information for better decision-making.

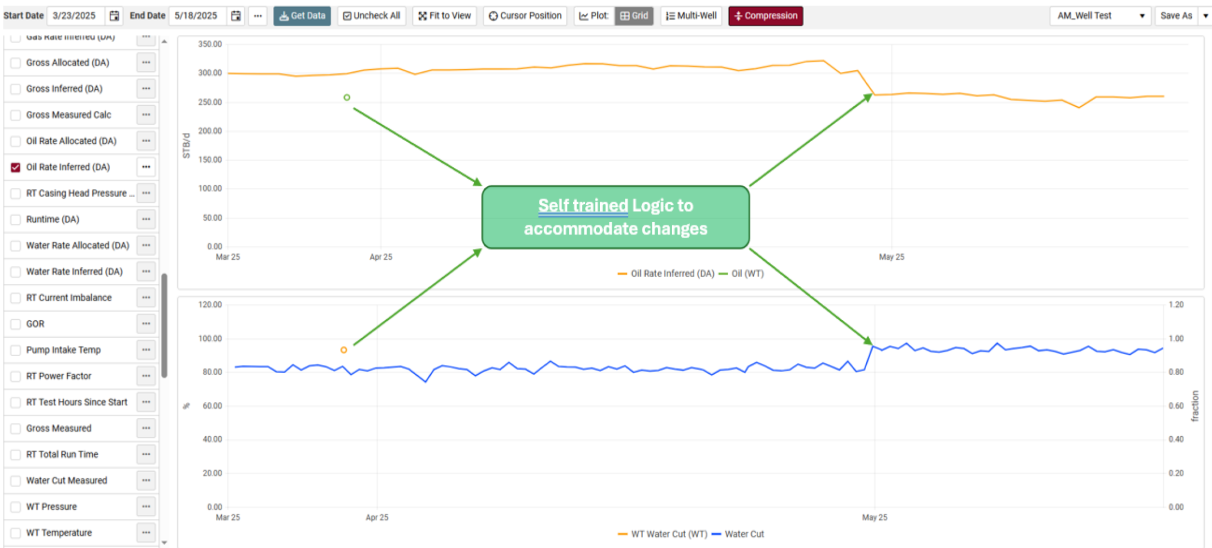


Figure 7—Case 4 - Well D

Validation and Results

In Figure 8 A comparison is shown between estimated and actual WC from tested wells, with most data points aligning closely, except in cases where discrepancies exceed 10%. Causes for these differences include inaccurate WC readings, sensor data errors, and gas presence in the flow. Estimating water cut often differs from real measurements due to various factors:

- Incorrect WC measurement
- Real-time data inaccuracies
- Gas interference in output

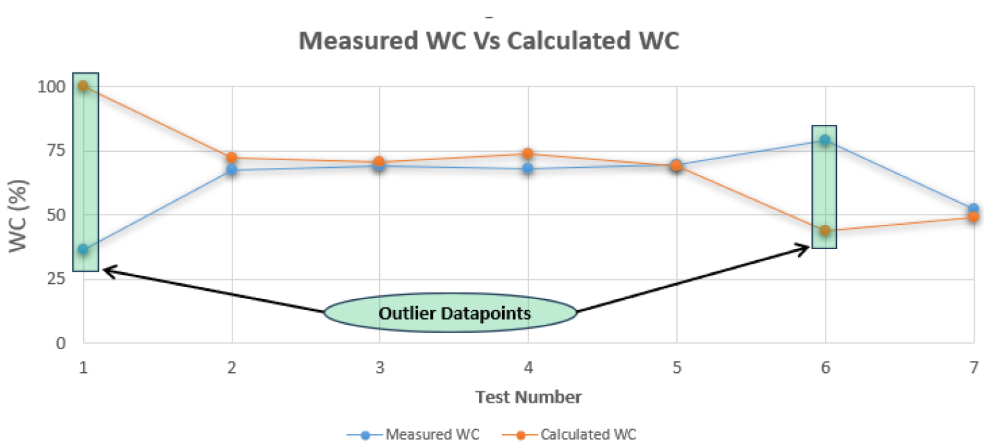
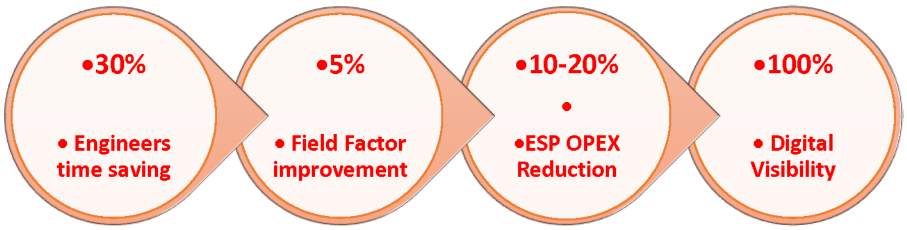


Figure 8—Outlier Datapoints for comparison.

Tracking oil shortfall proves time-intensive and challenging for production engineers when WC readings are off. However, integrating real-time logic for WC estimation has improved phase rate interpretation, reduced manual workload and boosted operational efficiency.

Improvement in the field factor has also been observed, which in turn supports the performance of physics-based well models, leading to better hydrocarbon allocation and enhanced production accounting.

Key Results



Integration of WC estimation logic with Well Management Visualization System (WMVS)

To effectively manage numerous wells and their frequent daily tests, the project team integrated the WC estimation logic into the internal WMVS platform. This integration connects various components such as the Well Model Engine, WMVS workflows, system-based validation rules, and APIs. These elements collectively address the constraints of manual calculations by enabling:

- Detection of anomalies in real-time sensor inputs
- Estimations across varied water cut levels—low, medium, and high
- Adaptation to emulsion effects and fluctuations in GOR
- Handling abrupt shifts in downhole pressure and well conditions caused by interference
- Real-time water cut prediction based on live data and physics-driven well rate models

This architecture not only automates but also enhances the reliability of water cut assessments, providing a scalable solution for field-wide operations.

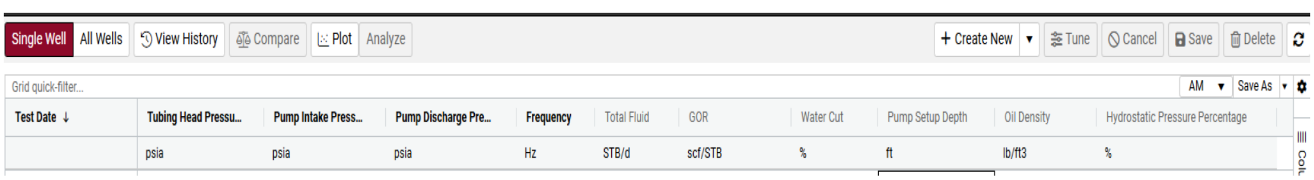


Figure 9—Water Cut estimation Page in Well Management Visualization System (Software User Manual).

Way Forward

- Project Team would be continuing to modify change management across OQEP asset to incorporate advisory modules along with automation of day-to-day calculation processes through DOF-WMVS. Team would be continuously working to change management across OQEP asset to incorporate advisory modules and automation of daily calculation processes using DOF- WMVS.
- OQEP is continuing to maximize ESP surveillance and optimization through the incorporation of science and technology into day-to-day operations.
- Workflow automation in well modeling and calculations has enabled integration of OQEP-DOF team data into a corporate database that has increased confidence in the quality and availability of data across E&P verticals, paving the way for ESP data analytics projects for failure prediction.
- Machine Learning (ML) and Artificial Intelligence (AI) models will be explored to:
    - Identify trends and anomalies in ESP performance.
    - Predict ESP failures with higher accuracy using historical and real-time data.

  - Recommend proactive maintenance actions, optimizing ESP run life and reducing unplanned shutdowns.
  - Enable closed-loop optimization by continuously learning from real-time feedback and adjusting operational setpoints automatically.
- The integration of AI/ML functionality into the existing WMVS infrastructure will deliver adaptive learning systems that evolve with field behavior, expanding on OQEP's vision for progressing towards fully autonomous ESP operations.
- These smart systems won't only enhance operational efficiency, they will also support strategic decision-making, resulting in a smarter, more data-driven E&P environment.

Change Management

On of the biggest hurdle to get asset-level acceptance of the technology and automation is the attitudes of the working engineers towards new systems and processes. This system has helped OQEP change in the working culture, introducing engineers to new methods of collaborative working. Training to engineers by DOF team was another way to cope with the innovative technology.

Conclusions

This method is extremely effective in predicting water cuts with a very high degree of confidence. As a result, of OQEP will be able to estimate and monitor water cuts in real-time, enabling better decision-making and optimizing production processes. The method was rigorously tested on wells with varying water cut values, showing excellent agreement between measured and computed water cuts, which further validates its reliability and accuracy.

One of the biggest hurdles to achieving asset-level acceptance of the technology and automation is overcoming the resistance of working engineers toward adopting new systems and processes. However, this system has significantly helped OQEP foster a positive shift in their working culture, introducing engineers to new, more efficient methods of collaborative working. The training provided by the DOF team was instrumental in helping engineers adapt to the innovative technology, ensuring they could fully leverage the new tools and systems. This approach not only facilitated the successful implementation of the system but also enhanced the engineers' skills and confidence in using advanced technologies for better operational outcomes.

Acknowledgements

The authors acknowledge the support of the management of OQ Exploration & Production LLC, Oman, for supporting this work and granting permission for this paper to be published. In addition, the authors acknowledge all other stakeholders from OQ Exploration & Production LLC and Weatherford Digital Solutions for their invaluable contribution in making this workflow deployment a success. Their dedication, expertise, and collaboration were essential in achieving the successful implementation and optimization of this project.

Nomenclature

AI = artificial intelligence
API = application programming interface
CAPEX = capital expenditure
DOF = digital oil field
E&P = exploration and production
ESP = electrical submersible pump
G_{AVG} = pressure gradient of producing liquid

HFSD = high frequency sensor data
IPR = inflow performance relationship
ML = machine learning
mMD = measured depth (meter)
mTVD = true vertical depth (meter)
OPEX = operating expenditure
OQEP = OQ Exploration and Production
P_f = pressure loss due to tubing friction
P_G = pressure due to height of liquid column
P_{PDP} = pump discharge pressure
P_{PIP} = pump intake pressure
P_{THP} = tubing head pressure
PVT Data = pressure, volume, temperature data
RTWC = real time water cut
SG_{AVG} = average specific gravity
SQL = structured query language
WC = water cut
WMS = well management system
WMVS = well management visualization system
VBA = Visual Basic Application

References

Brahmantiyo Aji Sumarto, Bogi Haryo Nugroho, Bibi Hussain Akbar et al 2022. Water Cut Estimation Using ESP Sensor Reading and Well Test Data: A Novel Approach. Presented at the *SPE ADIPEC*, Abu-Dhabi, UAE., 3 November. SPE-211407-MS. https://doi.org/10.2118/211407-MS

Kawther Al Jabri, Ahmed Al Hinai, Anand Raosaheb More et al 2024. Automated Production Allocation & Deferment Calculation Using Integrated Digital Oil Field Realtime Surveillance Data and Physics Based Well-Models. Presented at the *SPE ADIPEC*, Abu-Dhabi, UAE., SPE-222528-MS. https://doi.org/10.2118/222528-MS

Kawther Al Jabri, Ahmed Al Hinai, Anand Raosaheb More et al 2024. Logical Method to Estimate Water-Cut in Real Time for ESP-Wells Using Real-Time and Well Data. Presented at 10th Middle East Artificial Lift Conference, Muscat, Oman., MLF-81, 20 – 22 January 2025

Sumarto B.A., Nugroho B.H., Joenaedy M.A. et al 2024. Harnessing Data Science for Produced Water Estimation: Unleashing the Power of Data-Driven Techniques. Presented at the *SPE ADIPEC*, Abu-Dhabi, UAE., SPE-219016-MS. https://doi.org/10.2118/219016-MS

Table 2C – Friction Loss in A.P.I. Tubulars

Takacs, Gabor, A critical comparison of TDH calculation models in manual ESP design procedures, *Journal of Petroleum Science and Engineering*, December 2020

Takacs, Gabor, Electrical Submersible Pumps Manual Design, Operations and Maintenance, Second edition, Elsevier, 2018

Weatherford ForeSite® Software User Manual. 2024. Version 5.5 Release, 22nd April 2025.